

Input: Hydrometeorological Time Series

$\sim 8 \times 8 \text{ km}$ context; 72 hour lookback

Precipitation



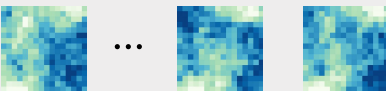
← Time

Soil Moisture



← Time

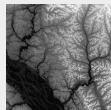
Snow Water Equivalent



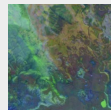
← Time

Spatial Context: Terrain and AlphaEarth Conditioning - Same $\sim 8 \times 8 \text{ km}$

[Height above nearest drainage, Topological Wetness Index, Global Curve Numbers (Dry, Average, Wet), POLARIS Soil Properties (ksat, clay, theta_s), Google AlphaEarth]



...



Conv2D Encoder ↓

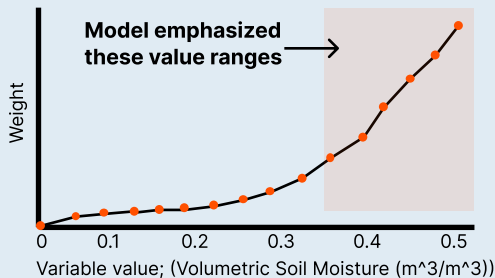
32 Parameters per variable

(16 control points: x_i, y_i)



Value Modulator

Learns a weighting function over each variable's raw value ranges. Runs variable-wise to normalize each variable to $[0,1]$



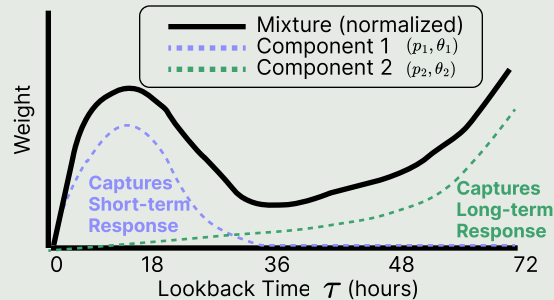
Conv2D Encoder ↓

5 Parameters per variable ($p_1, \theta_1, p_2, \theta_2, w$)



Temporal Modulator

Learns a temporal lookback weighting function over the 72 hour look back windows via a two-mixture Gamma kernel



→ **Modulated output**